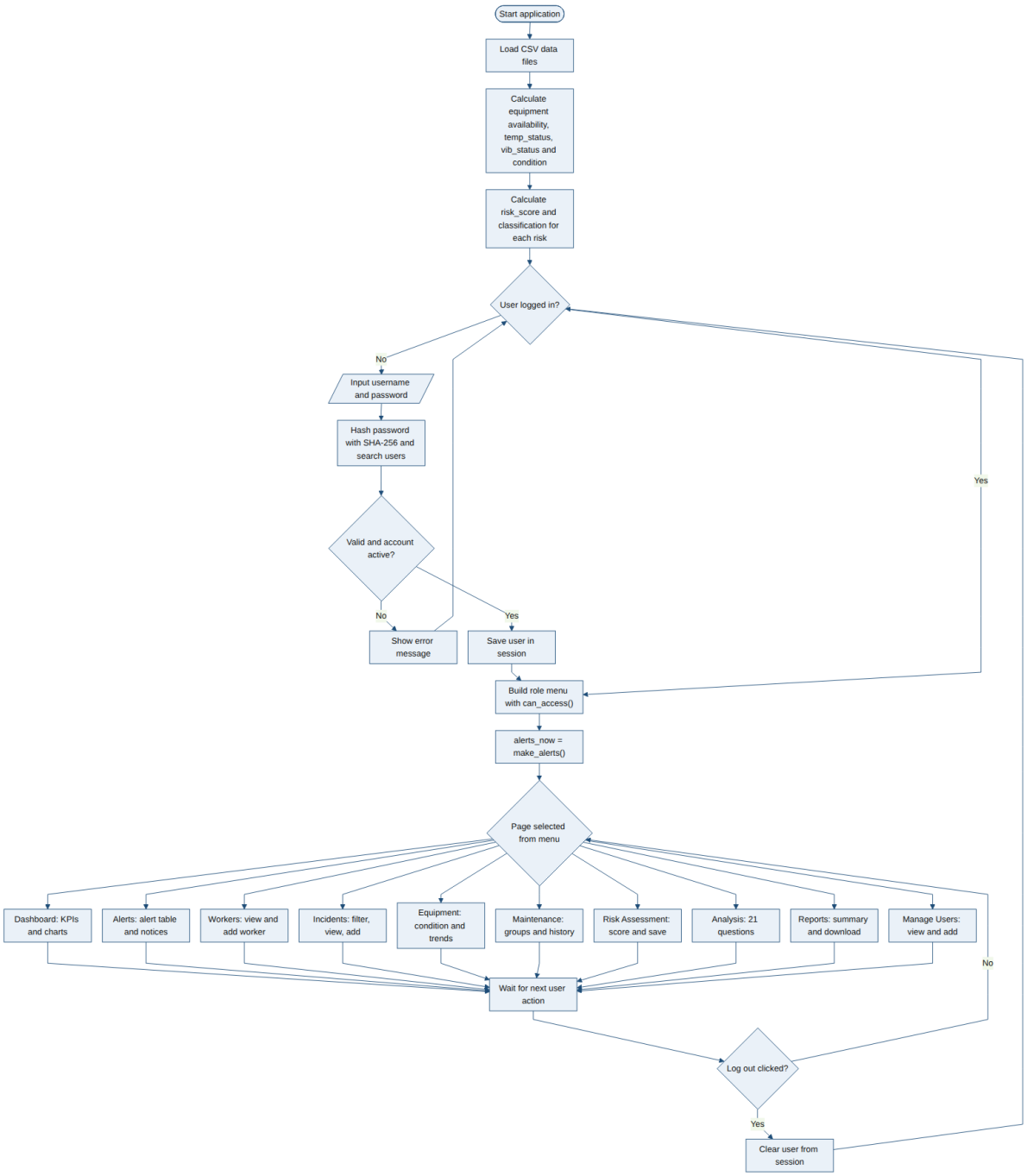
FlowCharts for Part a and b



Psuedocode

1.Start

2. Load and prepare data

LOAD users, permissions, workers, incidents, equipment, readings,

alerts_log, maintenance_log, risks FROM CSV files

```
FOR EACH row IN equipment
  availability <- get_availability(operating_time, downtime)
  temp_status  <- get_temp_status(temperature)
  vib_status   <- get_vib_status(vibration)
  IF temp_status = "CRITICAL" OR vib_status = "CRITICAL" THEN
    condition <- "CRITICAL"
  ELSE IF temp_status = "WARNING" OR vib_status = "WARNING" THEN
    condition <- "WARNING"
  ELSE
    condition <- "NORMAL"
  END IF
END FOR

FOR EACH row IN risks
  risk_score     <- likelihood * consequence
  classification <- get_risk_class(risk_score)
END FOR
```

3. Login and access control

```
IF no user in session THEN
  INPUT username, password
  IF "Log in" clicked THEN
    hashed <- SHA256(password)
    found  <- user whose username and password_hash match
    IF found = NONE THEN
      DISPLAY "Wrong username or password"
    ELSE IF found.active <> "Yes" THEN
      DISPLAY "This account is not active"
    ELSE
      store found in session
      reload page
    END IF
```

```
    END IF
    STOP
  END IF
  4. Menu and alerts
  role    <- user.role
  my_pages <- all pages where can_access(page rule) = True
  DISPLAY user name, role and my_pages as sidebar menu
  alerts_now <- make_alerts()

  5. Show the selected page
  page <- page chosen from my_pages
  CASE page OF
    "Dashboard"      : DISPLAY KPI metrics and charts
    "Alerts"         : DISPLAY alerts_now and equipment alert notices
    "Workers"        : DISPLAY workers and warnings; ADD worker with risk level
    "Incidents"      : FILTER and DISPLAY incidents; ADD incident
    "Equipment"      : DISPLAY condition, status messages and trend charts
    "Maintenance"    : DISPLAY maintenance groups and history
    "Risk Assessment" : CALCULATE score and class; SAVE assessment
    "Analysis"       : ANSWER selected question with table or chart
    "Reports"        : BUILD summary and recommendations; DOWNLOAD report
    "Manage Users"   : DISPLAY users; ADD user with hashed password
  END CASE
  6. Log out
  IF "Log out" clicked THEN
    clear user from session
    reload page (return to login)
  END IF

END PROGRAM
```